

Trader Behavior vs Market Sentiment

Introduction

This analysis explores the relationship between **trader behavior** and **market sentiment** using two datasets:

- **Bitcoin Market Sentiment dataset** → provides daily labels of *Fear*, *Greed*, *Extreme Fear*, *Extreme Greed*, and *Neutral*.
- **Historical Trader Data (Hyperliquid)** → contains trade-level details such as execution price, size, side (buy/sell), and closed profit & loss.

Objective: To understand how profitability, trade size, and volatility align or diverge from overall market sentiment.

2. Methodology

1. Data Cleaning

- Standardized timestamp columns into daily format.
- Removed irrelevant columns (IDs, hashes).

2. Aggregation

- For each day, calculated:
 - Average and total daily PnL.
 - Standard deviation of PnL (volatility).
 - Average and total trade size (USD).

3. Merge

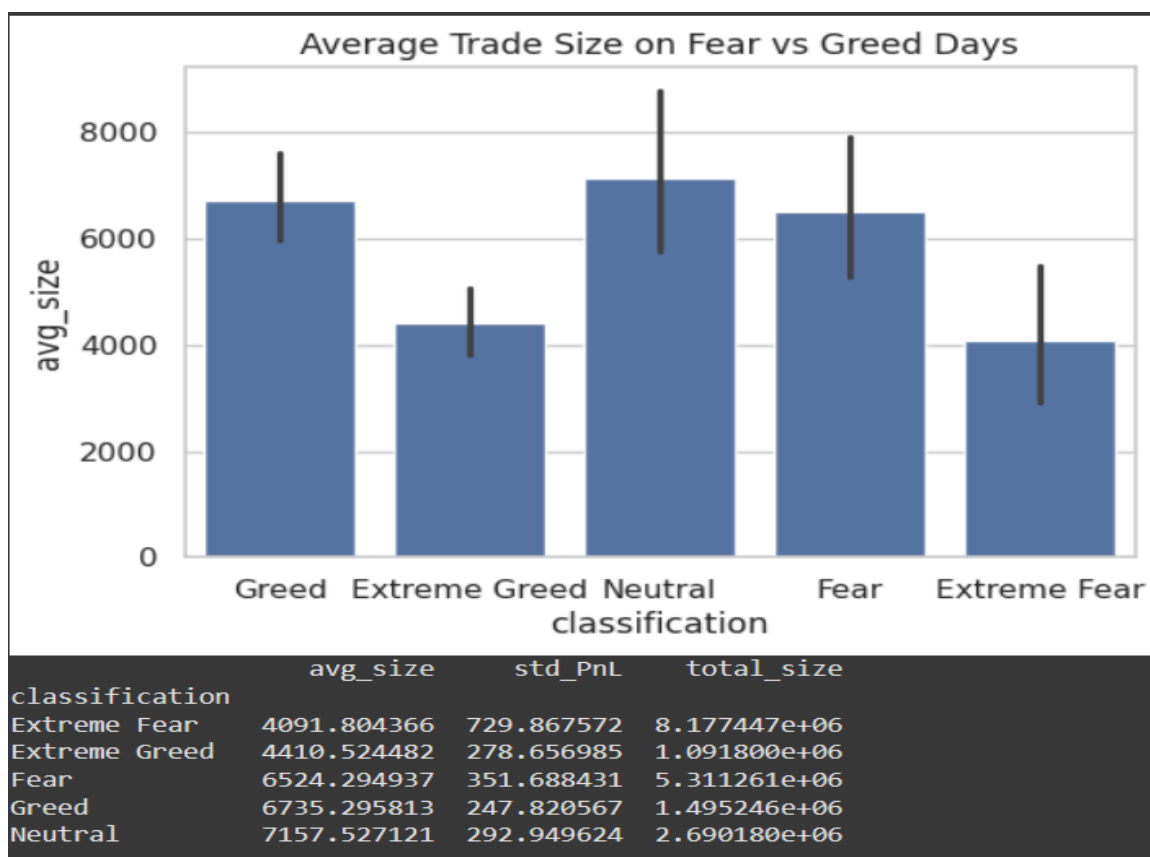
- Joined aggregated trader data with daily sentiment classification on **Date**.

4. Visualization

- Created comparative plots (Fear vs Greed categories).

3. Findings

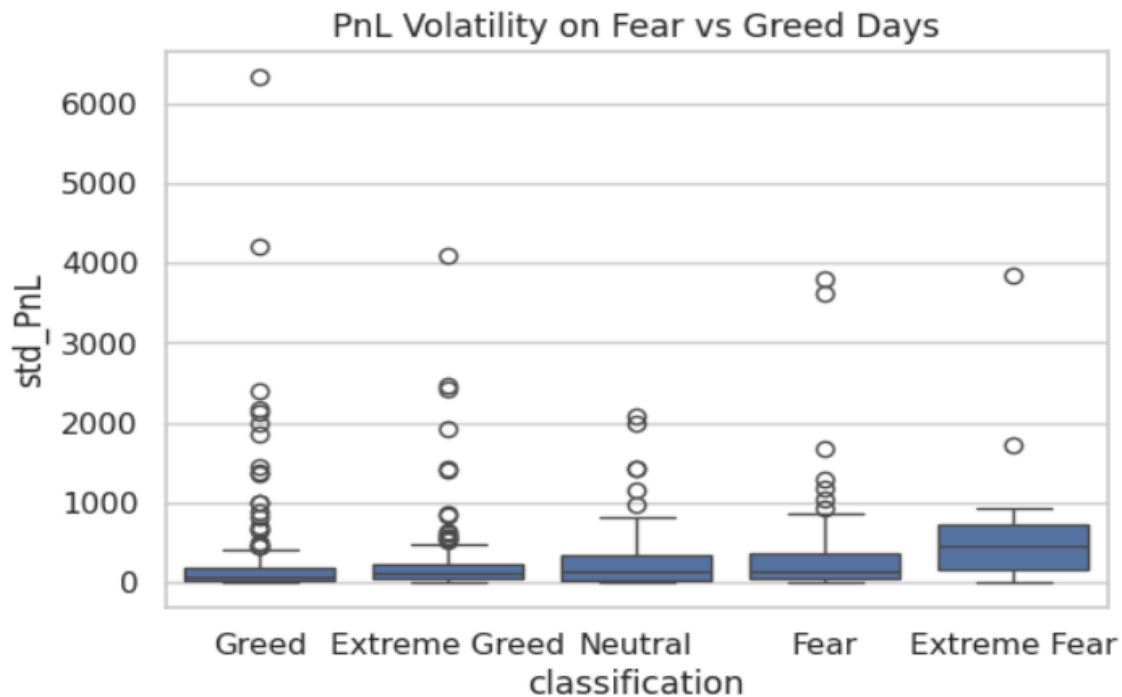
3.1 Average Trade Size on Fear vs Greed Days



- Traders tend to place **larger trades on Greed/Extreme Greed days**, indicating increased confidence and willingness to take bigger positions.

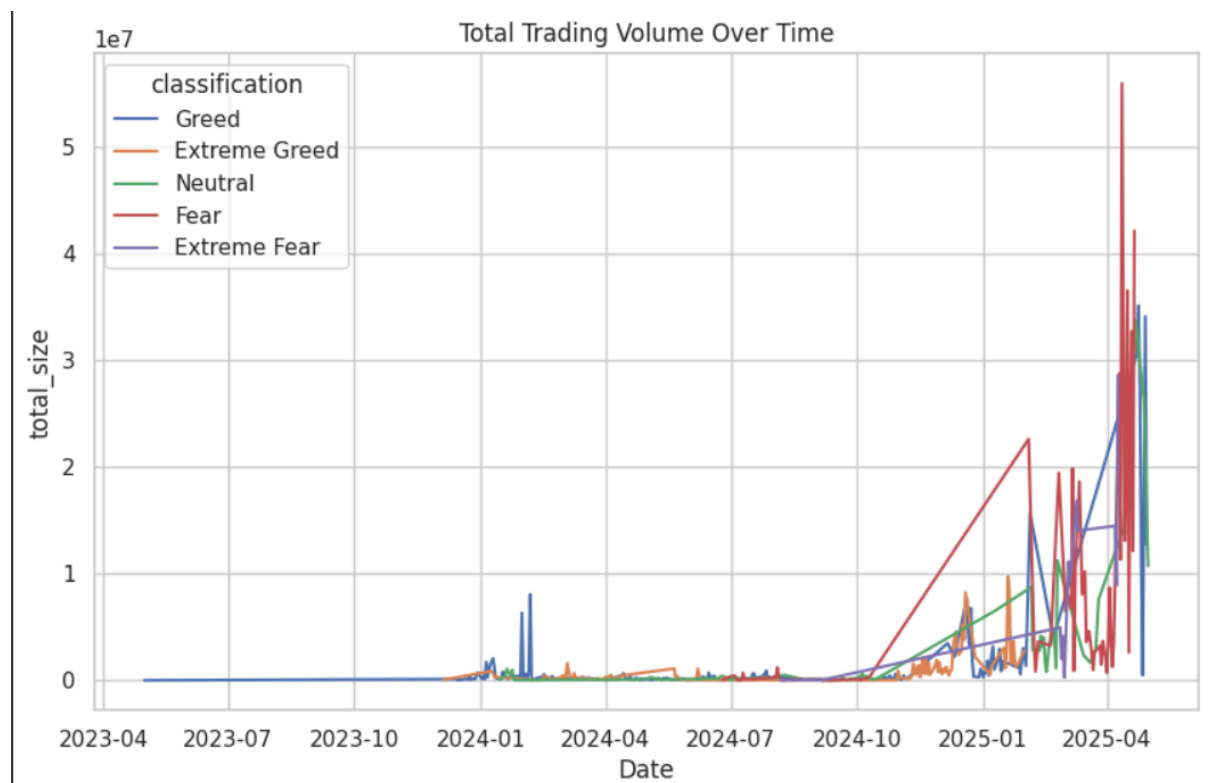
- On Fear days, average trade size decreases, showing cautious trading behavior.

3.2 PnL Volatility vs Sentiment



- Volatility of PnL is **highest on Extreme Fear days**, suggesting more erratic and unpredictable trading outcomes.
- Neutral days exhibit the **lowest volatility**, indicating stability.
- Surprisingly, Greed days show **less volatility than Fear days**, meaning fear drives more chaotic trading than greed.

3.3 Total Trading Volume Trend



- Trading volume spikes during **Extreme Greed days**, as traders enter markets aggressively.
- Extreme Fear days often see reduced total volume, though volatility per trade is higher.
- This indicates participation shrinks during fear, but those who trade behave more aggressively.

4. Conclusion

- **Greed sentiment** → larger trade sizes, higher total volume, but controlled volatility.
- **Fear sentiment** → reduced participation but **increased volatility**

in **PnL**, suggesting irrational or panic-driven trades.

- **Neutral sentiment** → most stable trading environment.

Takeaway: Understanding sentiment-driven patterns can help design strategies to **reduce risk on Fear days** and **capitalize on high volumes during Greed days**.